

Project Initialization and Planning Phase

Date	29/09/2026
Team ID	UNM1425EduGenie
Project Title	EduGenie
Maximum Marks	5 Marks

PROJECT PROPOSAL (PROPOSED SOLUTION) REPORT

The EduGenie project aims to provide students with an AI-powered personalized learning assistant. It addresses common learning challenges such as difficulty understanding complex concepts, finding immediate answers, preparing practice quizzes, summarizing lengthy content, and identifying suitable learning resources. EduGenie uses Generative AI to provide interactive and personalized educational assistance.

PROJECT OVERVIEW

Objective

The primary objective of EduGenie is to provide students with fast, simple, and personalized educational assistance using Generative AI. The system helps students understand concepts, clarify questions, practice through quizzes, summarize educational content, and receive learning recommendations.

Scope

The project covers the development of a web-based AI learning assistant that supports Question Answering, Topic Explanation, Quiz Generation, Content Summarization, and Personalized Learning Recommendations. The system uses a FastAPI backend and Google Gemini API to generate educational responses.

PROBLEM STATEMENT

Description

Students may face difficulties understanding complex topics, obtaining immediate answers, preparing practice questions, and summarizing lengthy educational content. Traditional learning resources may not always provide explanations according to individual learning levels.

Impact

These challenges can lead to confusion, reduced confidence, difficulty in revision, and less effective learning. Providing personalized AI-based educational assistance can make learning more accessible, interactive, and efficient.

PROPOSED SOLUTION

Approach

EduGenie uses Generative AI to process student questions, topics, and educational content and generate responses based on the selected learning activity and learning level. The system provides personalized educational support through a simple web interface.

Key Features

- AI-powered Question Answering
- Personalized Topic Explanation
- Automatic Quiz Generation
- Educational Content Summarization
- Personalized Learning Recommendations
- Google Gemini API Integration
- Student-friendly web interface
- FastAPI-based backend
- Copyable AI-generated results

RESOURCE REQUIREMENTS

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Standard computer/laptop with internet connection
Memory	RAM	Minimum 4 GB recommended
Storage	Project files, dependencies, and logs	Minimum 5 GB available storage
Software	Programming Language	Python
Framework	Backend Framework	FastAPI
Frontend	Client-Side Technologies	HTML, CSS, JavaScript
AI Service	Generative AI API	Google Gemini API
Server	Development Server	Uvicorn
Version Control	Source Code Management	Git / GitHub
Development Environment	IDE	Visual Studio Code
Data	User Input	Questions, topics, and educational content provided by students